3) Explain about performance Characteristics of Instruments in brief

Performance Characteristics of Instruments

Performance characteristics means the properties of an instrument which tell us how well an instrument measures a quantity.

They are mainly divided into two types:

1. Static Characteristics – used when the measured quantity is not changing.
2. Dynamic Characteristics – used when the measured quantity is changing with time.

Static Characteristics

Static characteristics which are used to measure unvarying process conditions

1. Accuracy:

- It tells how close the measured value is to the true value.
- Higher accuracy means the measurement is more correct.

2. Resolution:

- It is the smallest change that an instrument can measure.
- It tells how small a change the instrument can detect.

3. Precision:

- It tells how closely repeated readings are to each other.
- Same readings again and again means high precision.

4. Expected Value:

- It is the value that we expect to get from the measurement.
- It is usually the correct or calculated value.

5. Error:

- It is the difference between the true value and measured value.
- Less error means better measurement.

6. Sensitivity:

- It tells how much the output changes when the input changes.
- More output change for small input change means high sensitivity.

Dynamic Characteristics

Dynamic characteristics are used when the measured quantity changes with time.

1. Speed of Response:

- It tells how quickly the instrument responds to a change.
- A fast response means the instrument gives the reading quickly.

2. Dynamic Error:

- It is the difference between the true value and measured value when the value is changing.
- It occurs because the instrument cannot respond instantly.

3. Lag:

- Lag means the delay in the instrument's response.
- The instrument takes some time to show the changed value.

4. Fidelity:

- It tells how well an instrument follows changes in the measured quantity.
- High fidelity means it follows the changes correctly.

7) Explain in detail the different types of errors in measuring instruments.

Error in measurement

- Measurement is the process of comparing an unknown quantity with an accepted standard quantity.
- The measurement thus obtained is a quantitative measure of the so-called "true value" (since it is very difficult to define the true value, the term "expected value" is used).
- Error may be expressed either as absolute or as percentage of error.
- Absolute error may be defined as the difference between the expected value of the variable and the measured value of the variable

where
$$e = Y_n - X_n$$

e = absolute error

Y_n = expected value

X_n = measured value

Therefore, % Error = (Absolute value / Expected value) × 100

= $(e / Y_n) \times 100$

Therefore, % Error = $((Y_n - X_n) / Y_n) \times 100$

It is more frequently expressed as accuracy rather than error.

Therefore, $A = 1 - \left| (Y_n - X_n) / Y_n \right|$

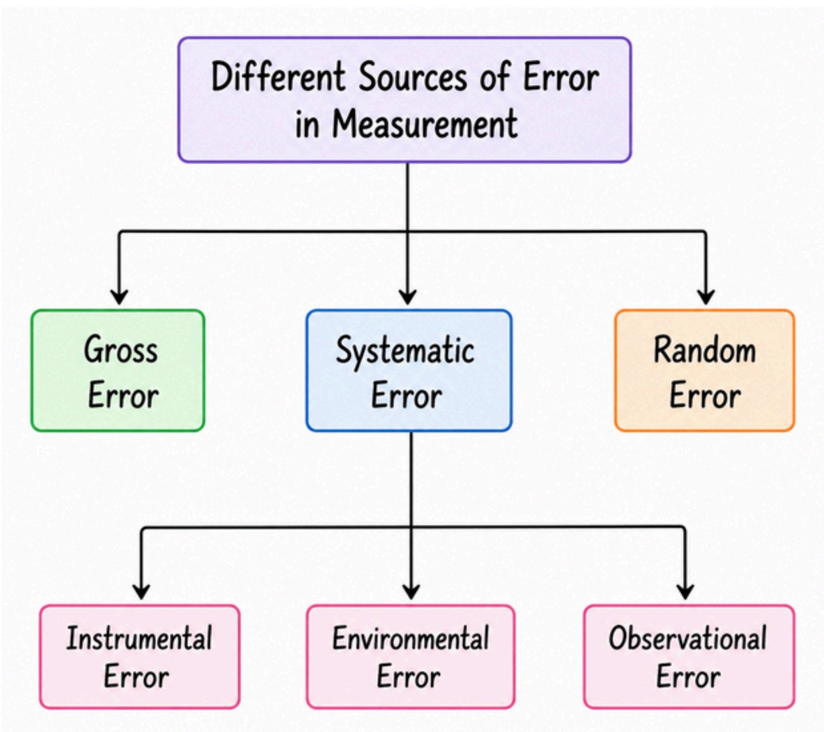
where A is the relative accuracy.

Accuracy is expressed as % accuracy.

a = 100% − % error

a = A × 100%

Different sources of errors in the measurement. These are classified as



1. Gross Error

- It occurs due to human mistakes while taking or recording readings.
- It may also occur due to wrong instrument adjustment or calculation mistakes.
- It can be reduced by taking and recording readings carefully.

2. Systematic Error

- It occurs due to instrument defects or environmental conditions.
- It gives an error in a fixed or predictable way.
- In general, systematic errors are classified as

(i) Instrumental error
(ii) Environmental error
(iii) Observational error

(i) Instrumental error :

→ It occurs due to defects or problems in the instrument.
→ Examples are friction, worn-out parts and spring problems.
→ It can be reduced by using a suitable and calibrated instrument.

(ii) Environmental error

→ It occurs due to changes in temperature, humidity, pressure, etc.
→ These external conditions affect the measurement.
→ It can be reduced by controlling the surrounding conditions.

(iii) Observational error

→ It occurs due to mistakes made by the observer while taking readings.
→ Common examples are parallax error and estimation error.
→ It can be reduced by taking the reading carefully

3.Random Error

- It occurs due to unknown and unpredictable causes.
- It is usually caused by the small effects that cannot be controlled.
- It can be reduced by taking more readings and finding the average.

8) Explain the Operation of dual slope integrating Type DVM

Dual-Slope Integrating DVM is a type of digital voltmeter.

- It measures the unknown input voltage by using an integrator and a reference voltage.
- The measurement is done in two stages, called two slopes.
- Finally, the result is counted and shown on the digital display.

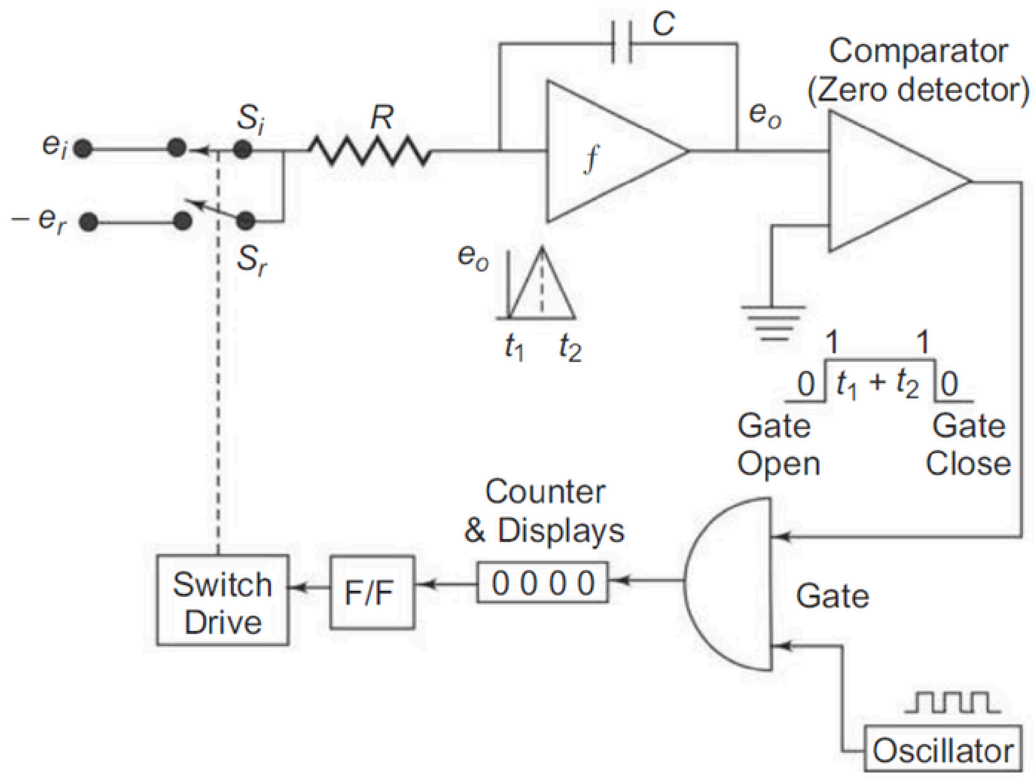


Fig. 5.4 Block diagram of a dual slope type DVM

Dual-Slope Integrating Type DVM – Working

The working is done in two stages or two slopes.

1. First Stage – Integration of unknown voltage

- First, switch Si is connected to the unknown input voltage Ei.
- The integrator integrates Ei for a fixed time t_1.
- So, the integrator output eo increases with a slope depending on the unknown voltage.
- Simple ga: Unknown voltage → Integrator → Output goes up

2. Second Stage – De-integration

- After time t_1, switch changes from Si to Sr and the reference voltage -Er is applied.
- Now the integrator output starts coming back towards zero.
- The time taken to reach zero is t_2.
- Simple ga: Reference voltage → Integrator → Output comes back to zero

3. Counting and Display

- During the time t_2, the oscillator gives clock pulses.
- These pulses are counted by the counter.
- The counter value is sent to the display, which shows the measured voltage.

Final Conclusion

Dual slope integrating DVM measures the input voltage by integrating it for a fixed time and then de-integrating using a reference voltage. The time taken to return to zero is proportional to the input voltage and is measured using a counter.

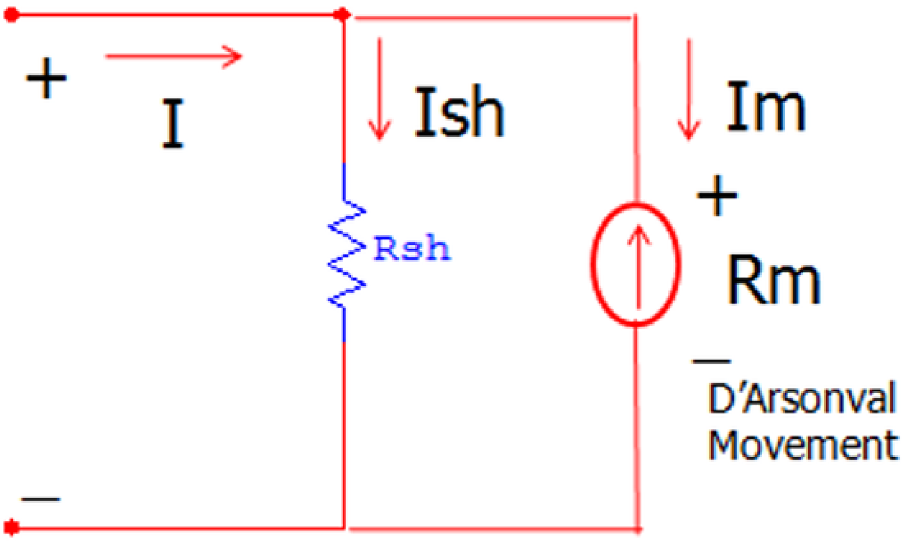
**Main point to remember:

Unknown voltage → Integrate → Reference voltage → Return to zero → Count time → Display voltage

6)Explain with a diagram how a PMMC can be used as an Ammeter

DC Ammeter

- The PMMC galvanometer constitutes the basic movement of a DC ammeter.
- The coil winding of the basic PMMC movement is small and light, so it can carry only a small current.
- To measure large currents, a low-value resistor called shunt resistor Rsh is connected in parallel with the PMMC movement.



Let,

R_m = Internal resistance of PMMC movement

R_{sh} = Shunt resistance

I_m = Full-scale deflection current of PMMC movement

I_{sh} = Current through shunt

I = Total current measured by ammeter

Since the PMMC movement and shunt resistance are connected in parallel, the voltage across both is equal.

Therefore, $I_m R_m = I_{sh} R_{sh}$

Also, $I = I_m + I_{sh}$

Therefore, $I_{sh} = I - I_m$

Substituting I_{sh} in the first equation, $I_m R_m = (I - I_m) R_{sh}$

Therefore, $R_{sh} = I_m R_m / (I - I_m)$

Hence, the required shunt resistance is: $R_{sh} = I_m R_m / (I - I_m)$

Thus, by connecting this low-value shunt resistance in parallel with the PMMC movement, the PMMC can be used as a DC ammeter.

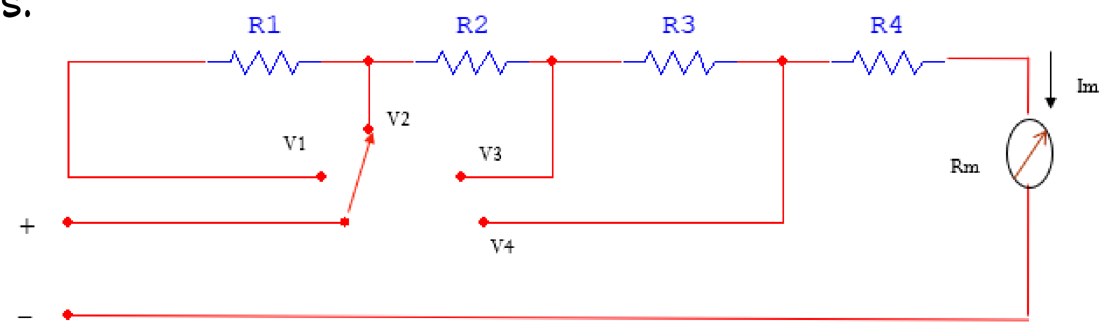
10) Explain the Construction and working of Multirange DC voltmeter

Multi-range DC Voltmeter

Introduction

A DC voltmeter can be converted into a multi-range voltmeter by connecting several resistors called multipliers in series with the meter movement.

It can measure different ranges of DC voltage by selecting different resistors.



Construction

- It consists of a PMMC meter movement having internal resistance Rm and full-scale current Im.
- Several resistors R1, R2, R3 and R4 are connected in series with the meter.
- A range selector switch is used to select the required voltage range V1, V2, V3 or V4.
- The selected resistors limit the current through the meter to a safe value.

Working

- When a DC voltage is applied, current flows through the selected resistors and the meter movement.
- For V1, only R1 + R2 + R3 + R4 are used, so the meter can measure the highest range.
- For V2, V3 and V4, different combinations of resistors are selected using the switch.
- The selected resistance controls the current so that the meter gives a full-scale reading at the selected voltage range.

Easy to remember

Different voltage → Select different resistors → Limit current → Meter shows reading

Limitations of Multi-Range Voltmeter

1. It has loading effect because the voltmeter draws some current from the circuit.
2. The accuracy is affected by the resistance of the meter movement and multiplier resistors.
3. Temperature changes can change the resistance values and cause errors.
4. The range selector switch may introduce contact resistance and measurement errors.
5. Applying a voltage higher than the selected range may damage the meter.

4)Explain the Successive approximation method in digital meters.

Successive Approximation DVM is a digital voltmeter in which the unknown input voltage is converted into digital form by successively comparing the input voltage with the output of a DAC.
It uses a Successive Approximation Register (SAR) to find the digital value of the input voltage.

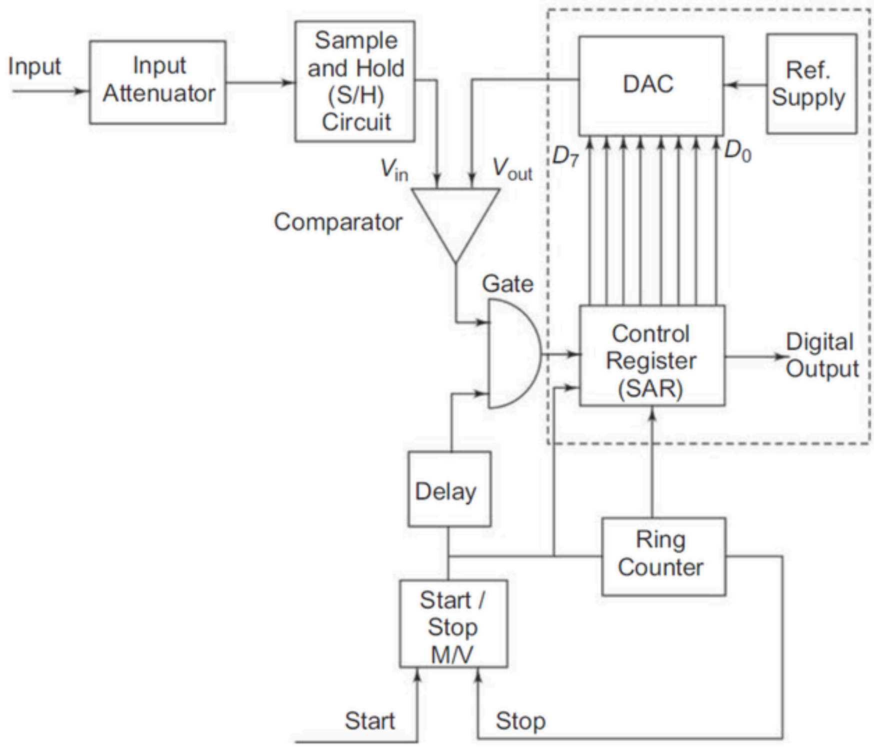


Fig. 5.10 Successive approximation DVM

Working

1. The unknown input voltage Vin is first applied to the input attenuator.
2. The Sample and Hold (S/H) circuit samples the input voltage and holds it constant during the conversion process.
3. The SAR initially sets the MSB D7 to 1 and all other bits to 0.
4. This digital code is given to the DAC. The DAC converts it into an equivalent analog voltage Vout.
5. The comparator compares Vin with Vout.
6. If Vin>Vout , the corresponding bit is retained (Set).

7. If Vin<Vout , the corresponding bit is reset.
8. The same process is repeated from MSB to LSB, one bit at a time.
9. Finally, all the accepted bits form the digital output corresponding to the input voltage.

$V_{in} = 1\,V$	Operation	D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0	Compare	Output	Voltage
00110011	D_7 Set	1	0	0	0	0	0	0	0	$V_{in} < V_{out}$	D_7 Reset	2.5
”	D_6 Set	0	1	0	0	0	0	0	0	$V_{in} < V_{out}$	D_6 Reset	1.25
”	D_5 Set	0	0	1	0	0	0	0	0	$V_{in} > V_{out}$	D_5 Set	0.625
”	D_4 Set	0	0	1	1	0	0	0	0	$V_{in} > V_{out}$	D_4 Set	0.9375
”	D_3 Set	0	0	1	1	1	0	0	0	$V_{in} < V_{out}$	D_3 Reset	0.9375
”	D_2 Set	0	0	1	1	0	1	0	0	$V_{in} < V_{out}$	D_2 Reset	0.9375
”	D_1 Set	0	0	1	1	0	0	1	0	$V_{in} > V_{out}$	D_1 Set	0.97725
”	D_0 Set	0	0	1	1	0	0	1	1	$V_{in} > V_{out}$	D_0 Set	0.99785

Thus, in a Successive Approximation DVM, the input voltage is compared successively with DAC-generated voltages, starting from the MSB and proceeding to the LSB, until the closest digital value of the input voltage is obtained.

1)Explain the working principle of digital frequency meter with a neat block diagram.

A digital frequency meter is an electronic instrument used to measure the frequency of an unknown periodic signal and display its value directly in digital form.
The basic principle of a digital frequency meter is to count the number of cycles of the unknown input signal occurring during a known and accurately controlled time interval.

$$f = N/T$$

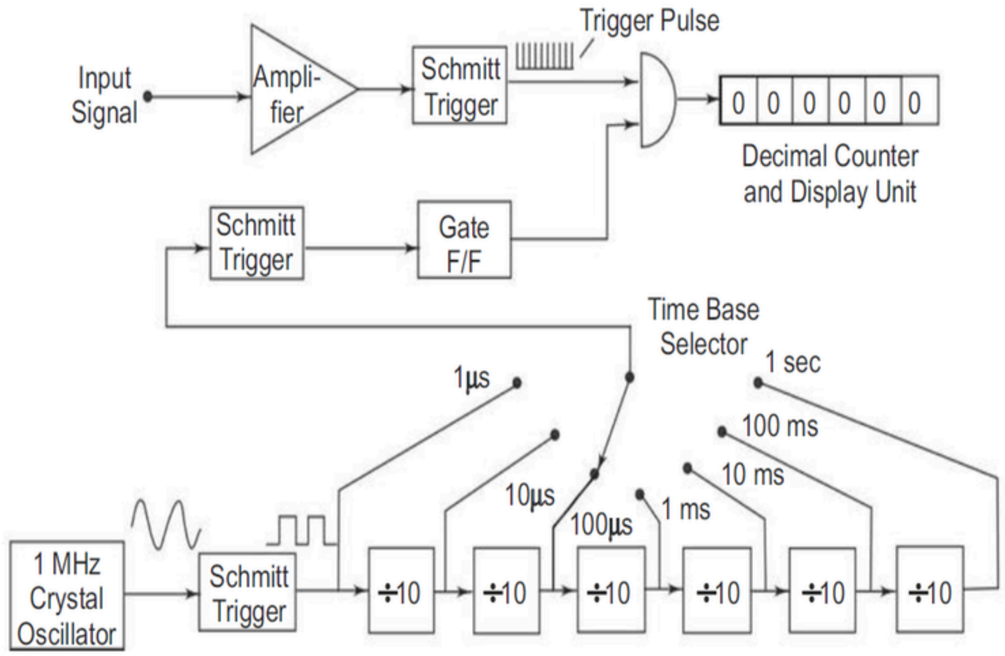


Fig. 6.7 Block diagram of a digital frequency meter

Working

- The unknown input signal is first given to the amplifier, where its amplitude is increased to a suitable level.
- The amplified signal is applied to the Schmitt Trigger.
- The Schmitt Trigger converts the input signal into a series of sharp rectangular trigger pulses.
- These pulses are applied to one input of the AND gate.
- A highly stable 1 MHz crystal oscillator generates the reference frequency.
- The crystal oscillator output is passed through Schmitt Trigger and frequency divider circuits to obtain accurate time intervals such as: 1μs,10μs,100μs,1ms,10ms,100ms,1s
- The required time interval is selected using the Time Base Selector.
- The selected time-base signal controls the Gate Flip-Flop (F/F). It opens the gate for a known time interval.
- During this gate-open period, the pulses from the unknown input signal pass through the AND gate and reach the decimal counter.
- The decimal counter counts the number of input pulses received during the selected gate time.
- The counted value is sent to the digital display, where the frequency is directly displayed.

2)Design a multi range DC Voltmeter with voltage ranges 0-10V, 0-50V, 0-100V, and
the meter movement having an internal resistance of 50 ohms and a full scale
deflection current of 2 mA.

Given:

Meter resistance,

$$R_m = 50\,\Omega$$

Full-scale deflection current,

$$I_m = 2\,mA = 0.002\,A$$

Voltage ranges:

$$V_1 = 10V, \qquad V_2 = 50V, \qquad V_3 = 100V$$

For a voltmeter,

$$R_T = \frac{V}{I_m}$$

and the required series resistance is

$$R_s = R_T - R_m$$

---

1. For 0–10 V range

$$R_{T1} = \frac{10}{0.002}$$

$$R_{T1} = 5000\,\Omega$$

Therefore,

$$R_1 = 5000 - 50$$

$$\boxed{R_1 = 4950\,\Omega}$$

2. For 0–50 V range

Total resistance required:

$$R_{T2} = \frac{50}{0.002}$$

$$R_{T2} = 25000\,\Omega$$

Therefore,

$$R_1 + R_2 = 25000 - 50$$

$$R_1 + R_2 = 24950\,\Omega$$

We already know:

$$R_1 = 4950\,\Omega$$

Hence,

$$R_2 = 24950 - 4950$$

$$\boxed{R_2 = 20000\,\Omega}$$

3. For 0–100 V range

Total resistance required:

$$R_{T3} = \frac{100}{0.002}$$

$$R_{T3} = 50000\ \Omega$$

Therefore,

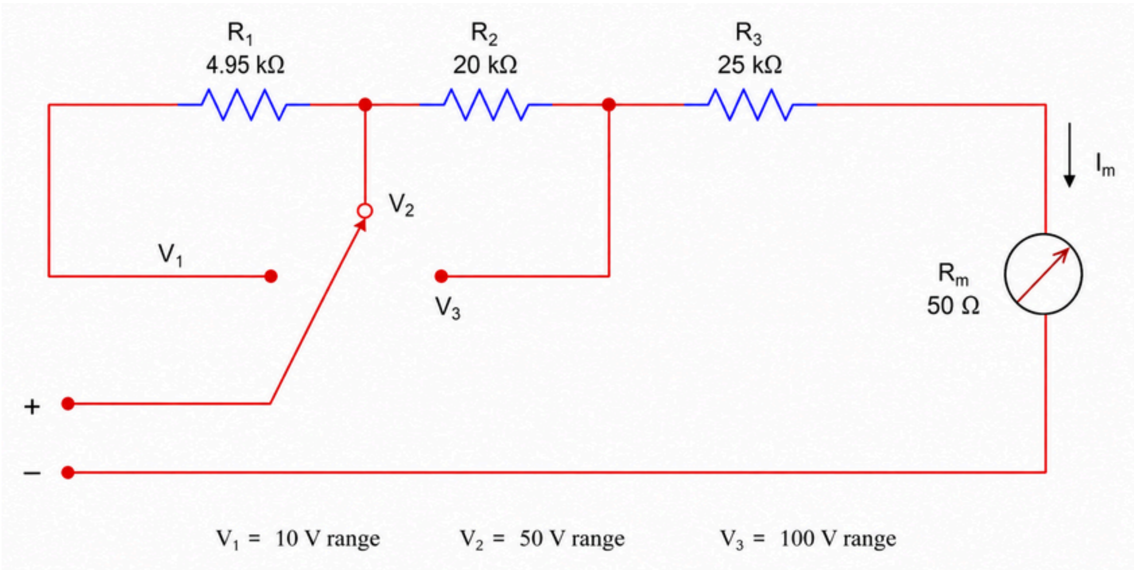
$$R_1 + R_2 + R_3 = 50000 - 50$$

$$R_1 + R_2 + R_3 = 49950\ \Omega$$

Hence,

$$R_3 = 49950 - (4950 + 20000)$$

$$\boxed{R_3 = 25000\ \Omega}$$



9)A 200 Ω basic movement is to be used as an ohmmeter requiring full scale deflection of 1 mA and internal battery voltage of 5 V. A half scale deflection marking of 2 kΩ is desired. Calculate:

i. The values of R1 and R2

ii. Maximum value of R to compensate for a 3% drop in battery voltage.

Given:

$$R_m = 200\Omega$$

$$I_{FSD} = 1mA = 0.001A$$

$$V = 5V$$

$$R_h = 2k\Omega = 2000\Omega$$

(i) Calculation of R_1

Formula:

$$R_1 = R_h - \frac{I_{FSD} R_m R_h}{V}$$

Substituting the values:

$$R_1 = 2000 - \frac{(0.001)(200)(2000)}{5}$$

$$R_1 = 2000 - \frac{400}{5}$$

$$R_1 = 2000 - 80$$

$$\boxed{R_1 = 1920\Omega}$$

Therefore,

$$\boxed{R_1 = 1.92k\Omega}$$

Calculation of R_2

Formula:

$$R_2 = \frac{I_{FSD} R_m R_h}{V - I_{FSD} R_h}$$

Substituting:

$$R_2 = \frac{(0.001)(200)(2000)}{5 - (0.001)(2000)}$$

$$R_2 = \frac{400}{5 - 2}$$

$$R_2 = \frac{400}{3}$$

$$\boxed{R_2 = 133.33\Omega}$$

(ii) Maximum value of R_2 for 3% battery drop

Initial battery voltage:

$$V = 5V$$

3% of battery voltage:

$$\frac{3}{100} \times 5 = 0.15V$$

Therefore, the new battery voltage is:

$$V' = 5 - 0.15$$

$$V' = 4.85V$$

For the reduced battery voltage:

$$R_{2(max)} = \frac{I_{FSD} R_m R_h}{V' - I_{FSD} R_h}$$

Substituting:

$$R_{2(max)} = \frac{(0.001)(200)(2000)}{4.85 - (0.001)(2000)}$$

$$R_{2(max)} = \frac{400}{4.85 - 2}$$

$$R_{2(max)} = \frac{400}{2.85}$$

$$\boxed{R_{2(max)} = 140.35\Omega}$$

Final Answer

$$\boxed{R_1 = 1.92k\Omega}$$

$$\boxed{R_2 = 133.33\Omega}$$

For a 3% drop in battery voltage:

$$\boxed{R_{2(max)} = 140.35\Omega}$$